

júliabiró

infrastructure & reliability engineer

about

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languages

fluent English
native Hungarian
B1 German

code

go python
javascript bash
Terraform, Kubernetes,
AWS, Fastly VCL,
nginx, everything
configuration

talks

 O'REILLY Velocity
 AWS Summit
 DevTalks Cluj

teaching

 django girls



interests

Reliability, scalability, security, monitoring, incident management

education

2007 – 2013 **BSc degree in IT Engineering/CS**
Budapest University of Technology and Economics
2002 – 2010 **MSc degree in Mathematics**
Eötvös Loránd University of Sciences

experience

2017 – 2025 **Contentful**
senior reliability engineer
2014 – 2017 **Prezi**
infrastructure and tooling engineer
2011 – 2014 **Ericsson Hungary**
software developer
2008 – 2011 **MTA SZTAKI (research institute)**
software developer

skills

Infrastructure management
AWS, Fastly, Kubernetes, Terraform
Monitoring
Prometheus, Grafana, Splunk, OpenTelemetry
Reliability
Networking, procedures
Caching
Fastly, Varnish, Akamai, CloudFront

scope

team
roadmap, project lead, implementation
group
collaboration on initiatives, establishing processes
across group limits
consulting on implementation
all engineering
establishing processes
across department limits
knowledge sharing, internal enablement

hobbies

Arthouse cinema, museums, quality journalism

projects

driver

Migrations

Contributed to the planning and execution of complex migration projects, ensuring zero-downtime transitions and data integrity. Migrated services between:

- Secret storage and access solutions
- Monitoring and alerting providers
- Application platforms, edge cache providers
- WAF solutions, async message bus systems
- Clouds (one small stateless service)

Monitoring

Participated in work in all layers and scopes of monitoring.

- Editing Grafana dashboards, managing log and metrics collection
- Managing monitoring data pipelines
- Led multiple rounds of restructuring alerting pipelines
- Defining future observability strategies

Reliability

This area is closest to my heart. I tend to contribute the reliability perspective to every project.

- Helped implement a multi-region setup for a stateful API
- Provided guidance on migrating traffic flows between async message bus systems
- Defined a process for managing reliability risks from a business perspective
- Conducted reliability and scalability reviews before high business-value events
- Led projects to reduce blast radius by segmenting and routing traffic

Incident management

I introduced and maintained incident management processes for an entire engineering org.

- Defined and documented processes, spread the culture of incident management internally
- I have been on-call for 10 years for customer-facing functionalities
- I actively participated in countless incidents, including customer incidents
- Participated in external communications about incidents

expert

Caching

I became local expert on our caching features.

- Deep experience debugging edge-layer issues, ie. connectivity and stale cache
- Enabled customer facing roles to handle caching-related issues and questions
- Planned roll-out of cache invalidation strategy changes to avoid stale data
- Improved internal and external product representation for caching-related features

Operations

Participated in the operation of application platforms for ten years.

- Managed dozens of Kubernetes clusters
- Managed a fleet of several hundred RabbitMQ instances
- Evaluated multi-region database backup solutions
- Maintained and developed both internal and external routing infrastructure

Architecture

My motto is that I prefer consistent mess over inconsistent mess.

- Mapped the state of internal routing and proposed future directions
- Suggested improvements for an organically diverse application platform for consistency and scalability
- Advised on projects involving data distributed among several infrastructure layers
- Promoted awareness to changes to failure domains and blast radiuses during project planning
- Assisted the adaptation of the infrastructure to a 100x increase in scale

experience

Security

I happened to sit close to security teams most of my career.

- Contributed to major changes of secret storage and access solutions
- Maintained the kube-secret-syncer
- Embedded representative of the Security Team, promoting security awareness
- Identified incidents where internal information was accidentally exposed publicly

Compliance

I participated in several external compliance projects.

- Defined and maintained Disaster Recovery Procedures, addressing both technical and customer-facing aspects
- Translated user access audit requirements for infrastructure tools and services to meet SOC2 compliance audit
- Assisted Business Continuity gameday

+1 Product-oriented

- Created structured definition of and scaled features that grew from one-off solutions to broadly marketed and deployed products
- Developed internal and external documentation, internal enablement, and managed business impact measurements for these features